AI-DRIVEN MULTI-ENVIRONMENT CINNAMON GROWTH MONITORING & AUTOMATED HARVESTING SYSTEM USING ROBOTIC MANIPULATION AND MULTISPECTRAL SENSING



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OUR TEAM

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INTRODUCTION



- CINNAMON IS ONE OF SRI LANKA’S MAJOR EXPORT CROPS
- TRADITIONAL CULTIVATION RELIES ON MANUAL EXPERIENCE
- ENVIRONMENTAL CONDITIONS STRONGLY AFFECT PLANT GROWTH
- HARVESTING PROCESS IS LABOR-INTENSIVE AND INEFFICIENT
- SMART TECHNOLOGIES CAN IMPROVE MONITORING AND PRODUCTIVITY

THE PROBLEM

KEY PROBLEMS:

- NO REAL-TIME MONITORING OF PLANT GROWTH
- ENVIRONMENTAL CONDITIONS VARY (HUMIDITY, LIGHT, SOIL MOISTURE)
- DISEASES ARE DETECTED TOO LATE
- FARMERS CANNOT PREDICT HARVEST READINESS
- CINNAMON BARK EXTRACTION IS MANUAL AND INEFFICIENT

IMPACT:

- LOW PRODUCTIVITY
- INCONSISTENT BARK QUALITY
- HIGH LABOUR COST



PROBLEM INTRODUCTION VIDEO

- BUILDING BLOCKS OF NATURE





USERS AND STAKEHOLDERS

USERS:

- CINNAMON FARMERS
- AGRICULTURAL RESEARCHERS
- PLANTATION MANAGERS

STAKEHOLDERS:

- CINNAMON EXPORT COMPANIES
- AGRICULTURAL TECHNOLOGY COMPANIES
- GOVERNMENT AGRICULTURE SECTOR

EXISTING SOLUTIONS & LIMITATIONS

EXISTING TECHNOLOGIES:

- MANUAL MONITORING
- BASIC IRRIGATION SYSTEMS
- SIMPLE PLANT DISEASE DETECTION TOOLS

LIMITATIONS:

- NO INTEGRATED SYSTEM
- NO PREDICTIVE GROWTH MODELS
- NO AUTOMATED HARVESTING
- NO ROBOTIC ASSISTANCE



PROPOSED SOLUTION

YOUR SYSTEM INTEGRATES:

- IOT ENVIRONMENTAL MONITORING
- MACHINE LEARNING GROWTH PREDICTION
- AI VISION GUIDED ROBOTIC HARVESTING

BENEFITS:

- REAL-TIME MONITORING
- PREDICT HARVEST TIME
- DETECT PLANT DISEASE EARLY
- AUTOMATE BARK HARVESTING



SYSTEM OVERVIEW





- IOT SENSORS MONITOR ENVIRONMENTAL CONDITIONS IN CINNAMON GROWTH CHAMBERS.
- SENSOR DATA IS TRANSMITTED TO A CLOUD PLATFORM FOR REAL-TIME MONITORING.
- MACHINE LEARNING MODELS ANALYZE THE COLLECTED DATA TO PREDICT PLANT GROWTH AND HARVEST READINESS.
- AI VISION SYSTEM DETECTS BARK BOUNDARIES AND DETERMINES THE OPTIMAL HARVESTING LOCATION.
- THE ROBOTIC ARM PERFORMS AUTOMATED CINNAMON BARK PEELING BASED ON AI GUIDANCE.

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MACHINE LEARNING-BASED GROWTH PREDICTION & ROBOTIC HARVEST GUIDANCE

- PREDICT PLANT GROWTH
- DETECT DISEASE
- ESTIMATE HARVEST READINESS



PROBLEM & OBJECTIVES

PROBLEM:

- FARMERS CANNOT PREDICT HARVEST READINESS.

OBJECTIVES:

- PREDICT PLANT GROWTH RATE
- DETECT DISEASE FROM IMAGES
- ESTIMATE BARK THICKNESS
- PROVIDE HARVEST GUIDANCE







RESEARCH METHODOLOGY

ALGORITHMS

- TIME-SERIES FORECASTING
- REGRESSION MODELS
- IMAGE CLASSIFICATION

TECHNOLOGIES

- PYTHON
- TENSORFLOW
- SCIKIT-LEARN
- OPENCV

VALIDATION

- MODEL ACCURACY
- PREDICTION ERROR
- CROSS-VALIDATION

IMPLEMENTATION

EXPLAIN:

- DATASET COLLECTION
- MODEL TRAINING
- PREDICTION SYSTEM
- ML OUTPUT SENT TO ROBOTIC ARM

OUTPUT:

ML-GUIDED HARVESTING SYSTEM

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IOT-DRIVEN MULTI-ENVIRONMENT MONITORING FOR ROBOTIC CINNAMON CULTIVATION

- MONITOR PLANT ENVIRONMENT CONDITIONS
- CONTROL GROWTH CHAMBERS
- SEND SENSOR DATA TO CLOUD DASHBOARD



PROBLEM & OBJECTIVES

PROBLEM:

- NO REAL-TIME ENVIRONMENTAL MONITORING FOR CINNAMON PLANTS

OBJECTIVES:

- MONITOR ENVIRONMENTAL CONDITIONS
- CONTROL GROWTH CHAMBERS
- SEND REAL-TIME SENSOR DATA
- INTEGRATE ROBOTIC ARM SENSORS







RESEARCH METHODOLOGY

ETHICS & SECURITY

- SECURE CLOUD DATA STORAGE
- CONTROLLED EXPERIMENT ENVIRONMENT

TECHNOLOGIES

- ARDUINO / RASPBERRY PI
- IOT SENSORS
- CLOUD DATABASE
- WEB DASHBOARD

VALIDATION

- COMPARE SENSOR DATA ACCURACY
- TEST ENVIRONMENTAL CONTROL SYSTEM

IMPLEMENTATION

EXPLAIN:

- GROWTH CHAMBER DESIGN
- SENSOR INSTALLATION
- IOT MICROCONTROLLER INTEGRATION
- ROBOTIC ARM SENSOR INTERFACE

OUTPUT:

IOT CONTROLLED MONITORING SYSTEM

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AI-GUIDED ROBOTIC ARM FOR AUTOMATED CINNAMON BARK HARVESTING

- AI VISION DETECTS BARK BOUNDARIES
- ROBOTIC ARM PERFORMS HARVESTING
- FULLY AUTOMATED PEELING PROCESS



PROBLEM & OBJECTIVES

PROBLEM:

- MANUAL CINNAMON BARK PEELING IS SLOW AND INCONSISTENT.

OBJECTIVES:

- DETECT HARVEST-READY BARK
- IDENTIFY PEELING BOUNDARIES
- CONTROL ROBOTIC ARM MOVEMENTS
- AUTOMATE HARVESTING PROCESS







RESEARCH METHODOLOGY

TECHNOLOGIES:

- COMPUTER VISION
- DEEP LEARNING
- OPENCV
- AI CONTROL ALGORITHMS

PROCESS:

- CAPTURE PLANT IMAGES
- DETECT BARK BOUNDARIES
- CALCULATE HARVESTING COORDINATES
- CONTROL ROBOTIC ARM MOVEMENTS

VALIDATION

- HARVEST ACCURACY
- PEELING QUALITY
- SYSTEM PERFORMANCE

IMPLEMENTATION

EXPLAIN:

- CAMERA INTEGRATION
- IMAGE PROCESSING
- ROBOTIC ARM CONTROL
- FULL SYSTEM INTEGRATION WITH IOT & ML

OUTPUT:

FULLY AUTOMATED AI HARVESTING SYSTEM

COMMERCIALIZATION PLAN

Target market:

- Cinnamon farmers
- Large plantations
- Agriculture technology companies

Revenue model:

- Sell smart agriculture system
- Hardware + software package

Value proposition:

- Increase productivity
- Reduce labour cost
- Improve cinnamon quality

COST & INVESTMENT

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COST

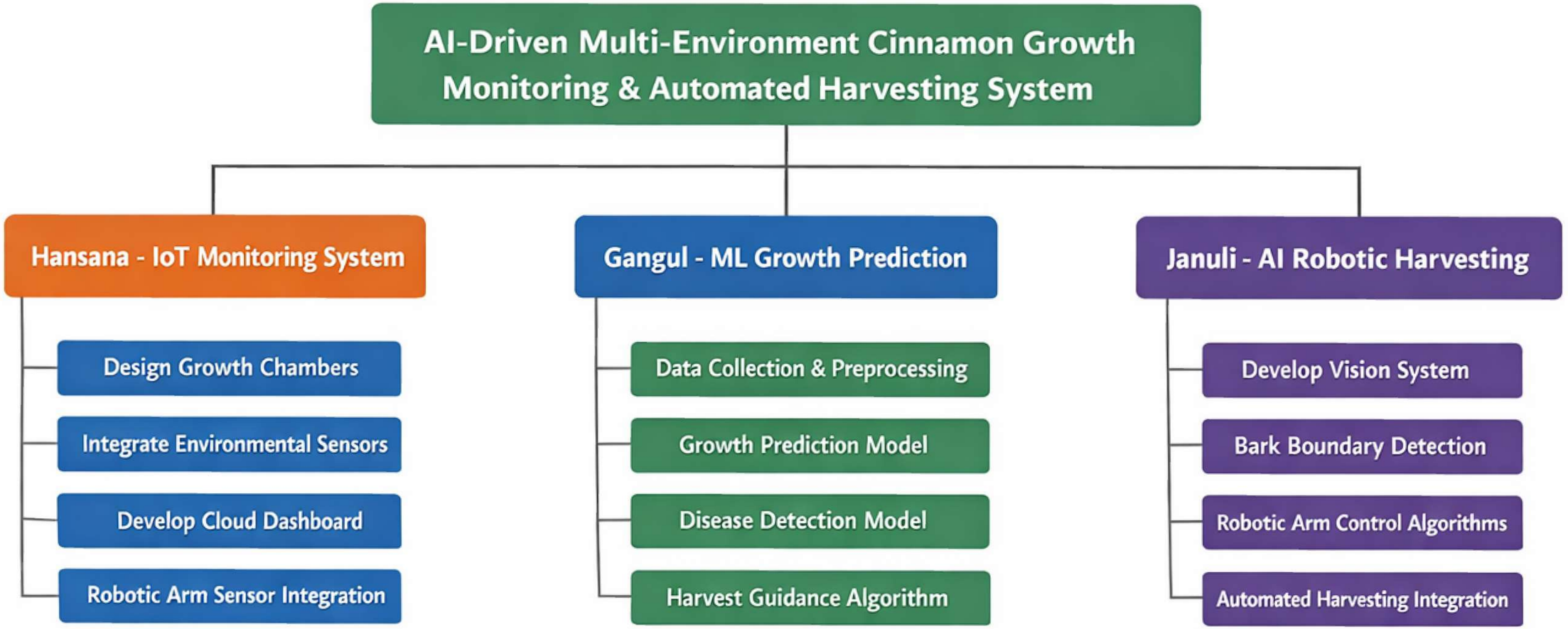
- Sensors
- IoT hardware
- Robotic arm
- Cloud services
- Development cos

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PRICING

- Sell system to plantations
- Maintenance subscription

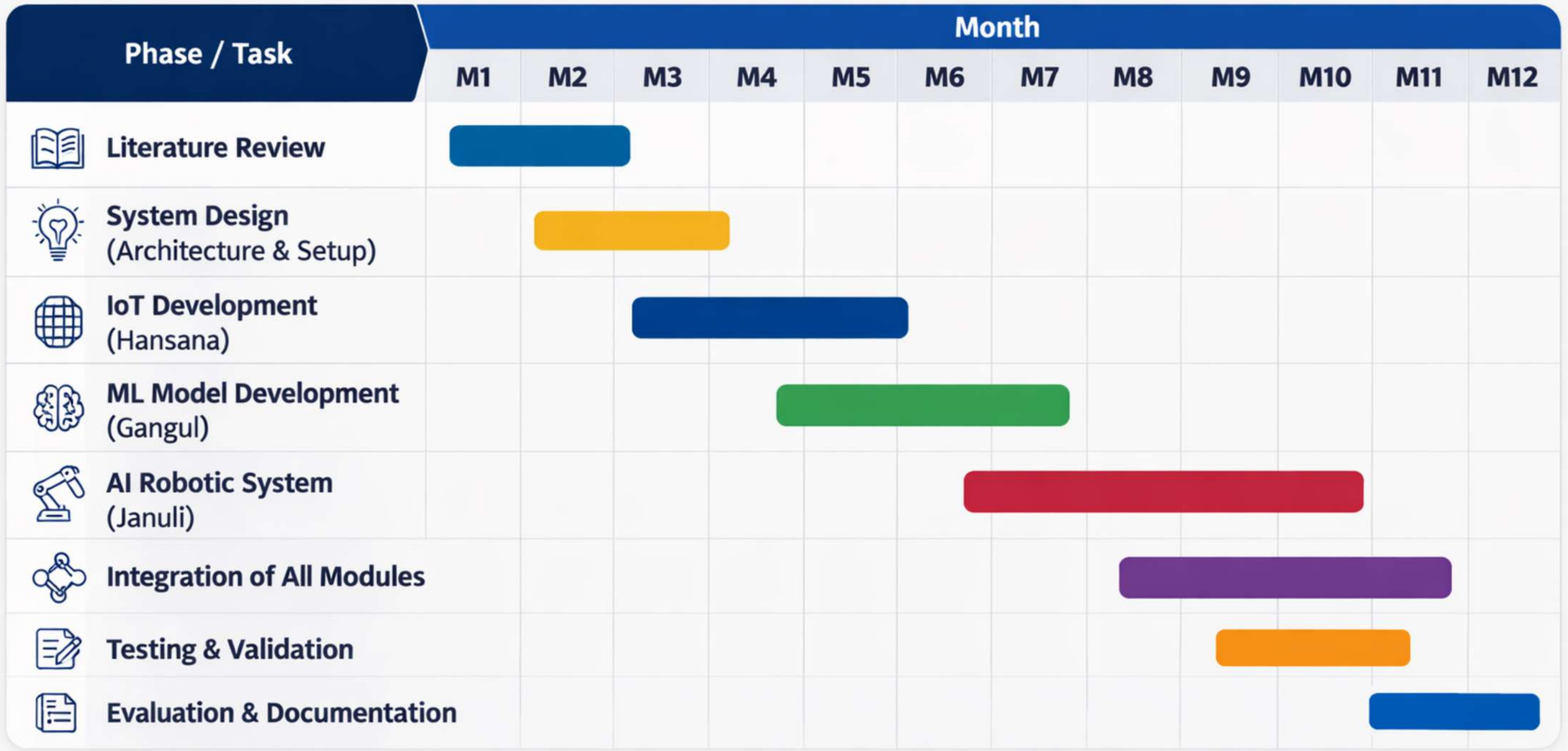
WORK BREAKDOWN STRUCTURE



GANTT CHART

AI-Driven Multi-Environment Cinnamon Growth Monitoring & Automated Harvesting System

Project Gantt Chart



THANK YOU

